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Report

On

“LUMISAT: MONITORING ARTIFICIAL LIGHT POLLUTION”

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ABSTRACT

LUMISAT is a CubeSat-based satellite designed to detect and monitor artificial light sources from a low-altitude platform. The satellite is engineered to analyse light pollution patterns and assess their environmental impacts by capturing light intensity, colour temperature, and spatial data from urban and rural zones. Equipped with multiple sensors and lightweight structural design using 3D-printed PLA, LUMISAT targets a flight altitude of approximately 1 kilometre. The project focuses on identifying sources of excessive artificial illumination that disturb ecological systems, affect human circadian rhythms, and contribute to energy-related carbon emissions. This project serves both as a technological demonstration and an environmental initiative to mitigate light pollution.

INTRODUCTION

The Rise of Aviation and Satellite Technology

Since the Wright brothers' first powered flight in 1903, the field of aviation has undergone revolutionary transformation. What began as rudimentary gliders has now evolved into an intricate network of high-speed jets, drones, and aerospace missions. Parallel to this growth, satellite technology, originally designed for military reconnaissance and weather monitoring, has expanded to include communications, navigation, Earth observation, and scientific research.

The launch of Sputnik 1 by the Soviet Union in 1957 marked the beginning of the satellite era. Since then, a myriad of satellites has been deployed in various orbits, enabling real-time weather tracking, GPS navigation, television broadcasting, global communications, and space exploration.

Miniaturization of electronics and the rise of commercial space entities have led to the development of CubeSats which are small, cost-effective satellites for research and innovation. These nanosatellites have become accessible to academic institutions, enabling student-led missions focused on diverse objectives such as climate monitoring, disaster response, and environmental science.

Evolution of CubeSats and Their Role in Environmental Monitoring

CubeSats typically range from 1U (10x10x10 cm) to larger modular forms such as 3U or 6U. Their compact form, standardized interface, and compatibility with launch vehicles have revolutionized remote sensing capabilities. Today, CubeSats serve as platforms for advanced payloads including spectrometers, cameras, GPS modules, and communication transceivers.

Environmental monitoring has become a prominent application of CubeSats. From tracking forest cover and glacier retreat to assessing ocean colour and air pollution, small satellites are now integral to sustainability research. With enhanced imaging resolution and sensor precision, even light pollution, an often overlooked but growing environmental issue—can now be detected from airborne or near-space platforms.

Understanding Light Pollution

Light pollution refers to the excessive or misdirected artificial light produced primarily in urban areas. While essential for human activity, artificial lighting has unintended consequences on ecosystems and human health. The effects include:

- Disruption of wildlife behaviour
- Human circadian rhythm disturbance
- Energy wastage and carbon emissions

Despite its widespread impact, light pollution remains under-monitored, especially in developing regions. Satellite-based monitoring provides a viable solution to assess and eventually regulate lighting practices.

The LUMISAT Initiative

Recognizing the growing need for data on artificial lighting and its consequences, the LUMISAT CubeSat was conceptualized and developed. LUMISAT aims to monitor artificial light sources at a low altitude of approximately 1 kilometre, making it suitable for high-resolution detection over specific geographic zones.

The CubeSat is equipped with the TCS34725 RGB colour sensor, a photodiode for ambient light measurement, MPU6050 motion sensor, BMP581 for temperature and pressure, powered by a rechargeable Li-Po battery regulated through a buck converter. LUMISAT's mission aligns with current efforts in environmental conservation, sustainable development, and smart urban planning. By mapping artificial light distributions, it supports policymakers, ecologists, and urban designers in making informed decisions.

LUMISAT Project Vision and Educational Purpose

LUMISAT is not merely a technical project but an educational project developed by undergraduate aerospace engineering students in order to have practical experience in the construction of CubeSats. The mission is to create awareness about the increasing problem of light pollution using available satellite technology. Every team member researched literature on at least five international journals, with this creating a profound depth of knowledge on the subject and enhancing teamwork skills. The satellite is constructed from low-cost, open-sourced parts such as Arduino and GPS modules to encourage educational access and innovation. Students in this project close the theoretical-practical gap and help address a globally applicable environmental concern.

LITERATURE REVIEW

CubeSats to measure light pollution

The research expands on previous studies that utilized DMSP and VIIRS satellites, which delivered valuable yet low-resolution information regarding artificial lighting. These shortcomings limited the ability to conduct an in-depth analysis of the effects of light pollution on wildlife, urban growth, and energy consumption. Recent studies emphasize CubeSats as an economical alternative, providing high-resolution nighttime images that are effective for identifying localized light sources. These capabilities enhance ecological connectivity modeling, urban development, and conservation initiatives by allowing for more accurate evaluations of human activities, habitat fragmentation, and the proliferation of artificial lighting.

Mapping Light Pollution, a CubeSat

Earlier research on light pollution mainly depended on satellites like DMSP and VIIRS, which provided data with low resolution and limited detailed analysis. These studies highlighted the harmful effects of artificial lighting on wildlife, ecosystems, and urban areas. With the progress in CubeSat technology, these compact satellites present a budget-friendly option for monitoring light pollution with high resolution. CubeSats allow for more precise mapping and observation of artificial light, enhancing environmental monitoring, conservation initiatives, and urban planning by delivering detailed insights into localized patterns of light pollution.

NITESat: A High Resolution, Full-Color, Light Pollution Imaging Satellite Mission

Earlier research on light pollution has relied on satellite technologies like DMSP and VIIRS, which have constrained resolution and are inadequate for comprehensive analysis. These studies highlight the impact of artificial lighting on ecosystems, the environment, and public health. The NITESat mission presents a high-resolution, full-color CubeSat specifically engineered to offer accurate imagery of light pollution. With its enhanced features, NITESat facilitates detailed mapping of artificial light, allowing for better environmental monitoring, conservation efforts, and urban planning by more effectively identifying localized patterns of light pollution.

CubeSat Nighttime Earth Observations

Research focused on nighttime Earth observations has depended on low-resolution satellite data from sources like DMSP and VIIRS, which did not provide sufficient detail for evaluating urban lighting and light pollution. The paper titled *"CubeSat Nighttime Earth Observations"* investigates the application of CubeSats with visible light cameras to capture high-resolution nighttime images. This method allows feasible to monitor usage of energy, environmental changes, and artificial lighting more precisely. CubeSats provide precise and accurate data, enhancing evaluations of urban growth, light pollution and the environmental impacts of human activity.

New Atmospheric Detector Technology Demonstrated on a CubeSat

In the past, the use of costly, big satellites for atmospheric monitoring has limited its flexibility and resolution. Recent developments in CubeSat technology provide atmospheric research an affordable substitute. The GRIFEX CubeSat mission, which used a readout integrated circuit and focal plane array to successfully test a high-speed atmospheric detector, is highlighted in the article "New Atmospheric Detector Technology Demonstrated on a CubeSat". This development has made it easier to conduct accurate, real-time atmospheric observations, creating new opportunities for Earth science missions that study air quality, pollution, and other atmospheric phenomena.

Device for automatic measurement of light pollution of the night sky

Manual observations and satellite imaging are two traditional light pollution monitoring techniques that frequently lack fine spatial resolution and real-time data. In recent years, automated mobile sensor networks have been developed that can measure the brightness of the night sky continually. In one approach, LoRaWAN communication technology is used to gather accurate, location-based data across urban regions. This innovation not only enhances light pollution tracking but also supports Smart City and IoT initiatives and offers a scalable, efficient method of managing artificial lighting and its environmental impacts in real time.

CubeSat System Structural Design

The research focuses on the creation of small, lightweight, and durable systems capable of enduring the extreme conditions of launch and space environments, particularly in the structural design of CubeSat computers. The aim is to facilitate integration by utilizing standardized CubeSat structures and aluminum alloys. The most commonly utilized technique for examining the impacts of vibration, stress, and temperature is finite element analysis, or FEA. The launch deployers and CubeSat Design Specification (CDS) ensures the structural compatibility. It goals to increase mass-to-strength ratios while maintaining durable during launch and in-orbit operations.

CubeSat Mission: From Design to Operation

From early concepts to practical deployment in orbit, the development and management of CubeSat missions necessitate a multidisciplinary approach. Challenges include managing orientation within a compact framework, communication restrictions, and constrained energy resources. The implementation of the CubeSat Design Specification has promoted global cooperation and standardization. Research highlights the importance of simulation, testing, and managing risks to enhance mission reliability. Improving autonomy and system integration continues to be a focus as CubeSats grow in various scientific, educational, and commercial space uses, showcasing their adaptability and cost-efficiency.

Mechanical design and finite element analysis of a 3 unit CubeSat structure

The aim of the CubeSat structural design study is to develop robust, lightweight systems that can experience the extreme conditions occurring in orbit. In order to identify the field of mechanical performance, it is essential to determine heat loads, vibrations, and stress accumulation regions. The other key method is finite element analysis (FEA). The researcher has investigated materials, design optimization, and customizable design to enhance structural reliability during the launch period and operation phases. CubeSat technology, also ensures stability and increase performance with more complex satellite systems, is being advanced by these studies, which additionally support the integration of CubeSats into larger space missions.

Design, Analysis, Optimization, Manufacturing, and Testing of a 2U CubeSat

Research on 2U CubeSats emphasizes the need to balance design efficiency with structural integrity to fulfill the requirements of space missions. Significant studies underline the application of Finite Element Analysis (FEA) to assess stress distribution, vibrational response, and thermal performance. Optimization methods are employed to lighten the structure while preserving strength and stability. Researchers also highlight the need of precise production and extensive testing in verifying functionality. Together, these results will help in the development of reliable, cost-effective CubeSat systems for challenging aerospace applications.

Optimization and Additive Manufacturing of a Single-Piece 1U CubeSat Primary Structure

Recent studies on CubeSat frameworks are increasingly concentrating on utilizing additive manufacturing to enhance design flexibility and production efficiency. Researchers focus on topology optimization to decrease weight while ensuring mechanical strength, particularly for compact 1U CubeSats. 3D printing permits single-piece frameworks with simplify assembly and improve structural integrity. Researcher also highlights the use of functional components in printed pieces, like hinges and locks. This method supports faster manufacturing cycles, reduce costs, and offers flexibility, making it ideal for the changing needs of small satellite missions.

A Case for a New Satellite Mission for Remote Sensing of Night Lights

This case study indicates the limitations of current satellite technology in adequately detecting and assessing artificial lighting at night. For precise monitoring of light pollution and its effects on the environment, astronomy, and urban growth, current methods are too sensitive and having low resolution. In order to enhance data quality, the authors recommend a new, specialized satellite mission, building on previous studies. In addition to filling important gaps in the current capabilities of remote sensing, such a mission would provide more effective scientific and policy applications.

Data analysis techniques in light pollution: A survey and taxonomy

The authors provide an objective review on the analytical techniques used to look into light pollution, showing their use in a range of research field. Techniques are categorized into key phases, including goal-setting, gathering data, preprocessing, and analysis, which offers a clear framework for comprehending the research process. The paper shows the merits and demerits of current techniques by examining a large number of previous studies. In addition to organizing current knowledge, the suggested taxonomy serves as a useful roadmap for upcoming studies, supporting methodological coherence and the creation of more complex analytical tools.

Spectral-level assessment of light pollution from urban façade lighting

This study examines the effects of urban facade lighting on light pollution while highlighting the spectrum characteristics of different light sources. It looks at how different wavelengths and intensities impact the nighttime environment in order to comprehend how light pollution affects both humans and biological organisms. This demonstrates the importance of looking at things at the spectral level. The goal of the study is to raise awareness of the issue of light pollution and encourage environmentally friendly urban lighting practices by examining the specific wavelengths that facade lighting emits.

Light pollution of freshwater ecosystems: principles, ecological impacts and remedies

This research investigates the impact of light pollution on freshwater ecosystems, looking into the ecological consequences and possible solutions. The authors emphasize how artificial lighting disrupts natural day-night and seasonal rhythms, adversely affecting the behavior, reproduction, and food webs of aquatic species. Light pollution can also change predator-prey relationships and interfere with migration patterns. The study proposes mitigation strategies such as utilizing environmentally friendly lighting technologies, shielding outdoor illumination, and enacting policies to curb light intrusion. These approaches aim to lessen the ecological effects of light pollution on freshwater environments and biodiversity.

OBJECTIVE OF THE PROPOSED SATELLITE WORK

Detection and Monitoring of Artificial Light Sources

LUMISAT's main goal is to detect and analyse artificial light sources from an airborne platform. The satellite will help map light pollution across many regions by gathering data on man-made light sources.

Because excessive artificial illumination disturbs ecosystems, interferes with human circadian rhythms, and indirectly adds to energy-related carbon emissions, this knowledge is essential for environmental planning. Keeping an eye on these emissions can aid in the formulation of well-informed plans to cut back on superfluous lighting and support sustainable urban growth.

Measurement of Light Intensity and Colour Temperature for Environmental Assessment

In addition to determining the total correlated colour temperature (CCT), LUMISAT will examine the intensity and spectral composition (red, green, and blue components) of the light within its field of view. The sources of cooler, high-intensity (blue-toned) lighting, which are more detrimental to wildlife and atmospheric conditions, will be identified with the use of this data.

Knowing these factors makes it easier to evaluate how artificial light affects the environment and climate change, which supports initiatives to minimize light pollution, improve lighting design, and lessen the consequences of global warming by promoting warm-spectrum, energy-efficient lighting.

Lightweight Structural Design for Efficient Deployment

Through 3D printing, the structural framework of the satellite is designed with lightweight materials like PLA, guaranteeing the best possible mass-to-function efficiency. In order to improve mission sustainability and lower launch costs, a low-mass design is essential.

Additionally, in small satellite missions like CubeSats, where weight and volume are strictly limited, lightweight constructions improve performance and increase payload capacity.

Ground Vision Capability for Surface-Level Observation

With its onboard image modules and sensors, LUMISAT provides a straightforward nadir (downward-facing) perspective, allowing for high-resolution observations of the Earth's surface from its assigned altitude.

The capacity to track localized light emissions with spatial accuracy and correlate them with particular urban or rural zones requires this "ground vision" capability. Such thorough field data helps environmental researchers, conservationists, and urban planners make data-driven judgments.

Target Altitude of 1 Kilo-meter for Focused Low-Altitude Remote Sensing

The satellite's operating height of roughly 1 km makes it perfect for low-altitude, high-resolution remote sensing applications. While reducing ambient distortion, operating in this range guarantees accurate detection of light intensity and colour fluctuations.

This targeted altitude range provides a scalable model for upcoming aerial light pollution monitoring systems, bridging the gap between high-orbit satellites and terrestrial sensors.

METHEDOLOGY

Structural Design, Simulation, and Testing

The LUMISAT CubeSat's structural design was created after a thorough literature research of material properties, CubeSat mission limitations, and tiny satellite geometries.

With dimensions of 10 cm x 10 cm x 15 cm, the finished design fits within the 1.5U CubeSat form factor.

- **CAD Modelling and Assembly**
SolidWorks and Fusion 360 were used to digitally model the structure, enabling high-precision 3D modelling, component placement, and interface verification. These technologies made it easier to iteratively improve the design in order to guarantee the best possible volume utilization, symmetry, and component accessibility.

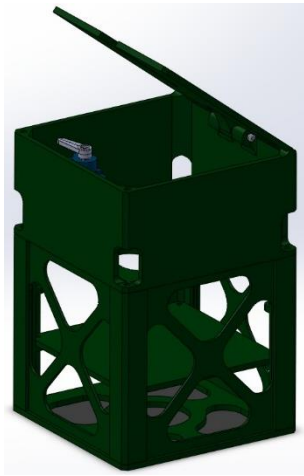


Figure 1: Structural Model

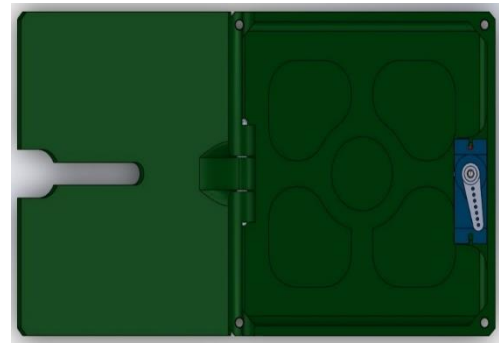


Figure 2: Parachute Bay

- **Selecting Materials and Finite Element Analysis (FEA)**

Because of its lightweight nature, biodegradability, and suitability for quick prototyping through 3D printing, polylactic acid (PLA) was chosen as the structural material. PLA can be used for short-duration missions and experimental CubeSat models, but it is not space-grade. Through FEA, structural analysis was carried out using ANSYS Workbench.

Through this simulation, the CubeSat's structural reaction was evaluated under several loading scenarios.



- **Simulation of Drop Testing**

ANSYS was used to simulate impact scenarios in which the satellite might experience abrupt deceleration or shock (e.g., during landing or unintentional drop) in order to conduct the drop test. The bottom surface of the structure experienced an impact velocity of 10 m/s from the simulation. By shedding light on possible stress concentrations, deformation patterns, and failure thresholds, the findings made sure that the design would still be structurally sound even in the event of small unintentional shocks.

Structural Mass Estimation and Optimization

Calculating the structural mass of the CubeSat with the slicing software Ultimaker Cura. The simulation was done to determine the weight of each 3D-printed component, not including the PCB plate and parachute material, to keep the overall structure within payload limits for launch and deployment.

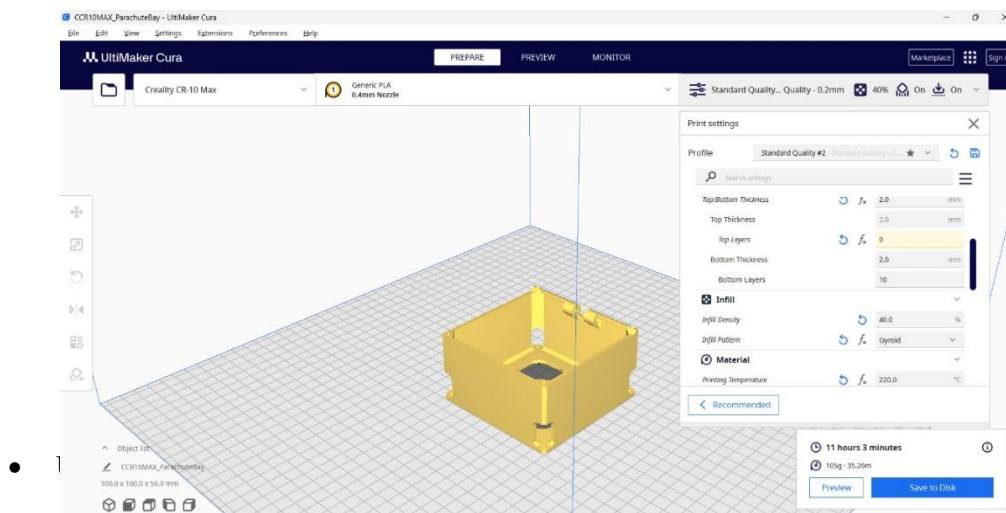
- Mass Estimation with Cura

Every structural piece was designed in Fusion 360 and saved as STL files in Cura. PLA was used as the chosen material, and the infill pattern of choice for internal structure was the gyroid pattern based on its excellent strength-to-weight ratio and load-isotropic properties. Cura calculated each component's mass as per 3D printing parameters like infill density, wall thickness, and support structures.

The breakdown of the component-wise weight is presented below:

Component	Weight (g)	Quantity	Total Mass (g)
Column	11 g	4	44 g
Side Wall	7 g	4	28 g
Parachute Bay	105 g	1	105 g
Parachute Plate	36 g	1	36 g
Hinge	1 g	1	1 g
Base Plate	31 g	1	31 g
Camera Plate	24 g	1	24 g
Total Structural Weight			269 g

This mass does **not** include the **electronic PCB** or the **parachute fabric**, which are evaluated in separate steps.



For all structural parts, a gyroid infill pattern was chosen in Cura because of its special characteristics:

Isotropic mechanical strength: Gyroid structures have equal strength in all directions, minimizing structural failure due to one-sided stress.

Lightweight structure: The continuous, curved geometry of the gyroid makes low material consumption with rigidity possible.

Damping of vibrations: Gyroid infill is useful in the absorption of vibrations and thus in aerospace applications where components are subject to launch dynamics and aerodynamic loads.

Optimized load distribution: Gyroid structures distribute mechanical loads evenly without developing stress concentration points.

Parachute Design and Descent Rate

- Parachute Design Configuration

To ensure safe recovery of the CubeSat structure during testing and descent operations, a passive parachute recovery system was designed. The parachute has the following characteristics:

Canopy Diameter	1.2 meters
Material	Ripstop Nylon (approx. 50GSM)
Suspension Lines	8 Nylon threads (120 cm long each)

Threads are paired into 4 groups, with each pair connected to one of the 4 structural mounting holes located on the satellite's columns

Even load distribution through symmetric attachment geometry to ensure stability during descent.

ANSYS Fluent was used to simulate the airflow interaction with the canopy form in a constant descent scenario in order to estimate the parachute's drag coefficient (C_d). The actual terminal velocity was then measured via physical test.

The two findings agreed on a drag coefficient of roughly C_d equals 0.9

The calculated terminal descent velocity is approximately 2.5 m/s, which ensures a safe landing speed for the CubeSat without damaging internal components. (Below Calculations provided)

The use of 8 suspension lines and symmetrical attachment provides dynamic stability during descent. This design approach balances lightweight construction, cost-effectiveness, and sufficient drag generation for controlled recovery.

Descent Rate Calculation

To estimate the **terminal velocity** (V_t) or **rate of descent**, we use the drag force equilibrium equation:

$$mg = \frac{1}{2}\rho AC_d V_t^2$$

Where:

- m = mass of the satellite = **0.4 kg**
- g = gravitational acceleration = **9.81 m/s²**
- ρ = air density at sea level = **1.225 kg/m³**
- A = reference area = projected area of parachute = $\pi \times \left(\frac{D}{2}\right)^2 = \pi \times (0.6)^2 \approx 1.131 \text{ m}^2$
- C_d = drag coefficient = **0.9**

Substituting values:

$$0.4 \times 9.81 = \frac{1}{2} \times 1.225 \times 1.131 \times 0.9 \times V_t^2$$

$$3.924 = 0.622 \times V_t^2$$

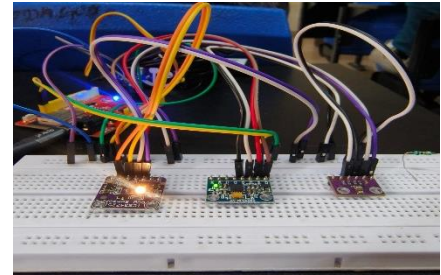
$$V_t^2 = \frac{3.924}{0.622} \approx 6.31$$

$$V_t \approx \sqrt{6.31} \approx \boxed{2.51 \text{ m/s}}$$

PCB Development and Sensor Compatibility Testing

- Preliminary Breadboard Prototyping

To create a preliminary prototype of the sensor layout and wire architecture, all of the onboard sensors were first organized and connected on a breadboard before the printed circuit board (PCB) design was finalized. During this phase, space limitations were visualized and component placement was optimized for effective routing on the finished PCB.



- Sensor Compatibility Testing with Bharat Pi Microcontroller

The microcontroller selected for the mission is Bharat Pi, chosen for its compact form factor and sufficient processing capability for low-orbit CubeSat missions. Each sensor was individually connected to the Bharat Pi to:

Verify electrical compatibility (voltage levels, I2C/SPI/UART protocol support.)

Test data acquisition integrity.

Confirm library/module support and communication stability in microcontroller firmware.

- Collective Integration and Code Validation

In order to evaluate the dependability of bus connectivity and multi-sensor data transfer, all sensors were linked simultaneously after individual validation. A single script for transmission was created in order to:

Get coordinated readings from every sensor.

Put the data in a structured format (such as CSV)

Use the specified communication interface to simulate transmission.

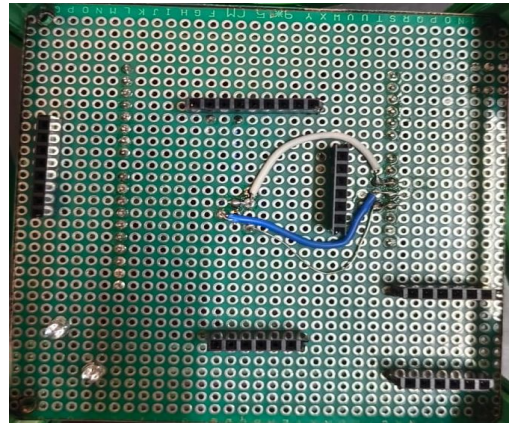
The microcontroller's ability to manage many sensor data inputs without experiencing data loss or protocol conflicts was confirmed by this collective code test.

- Final PCB Layout and Fabrication

An 8 cm by 8 cm double-layer PCB was created using the verified breadboard design as a guide. The double-layer arrangement was selected to: Minimize signal noise and improve routing. Differing ground and power planes to improve EMI control.

Make sure the compact integration fits the form factor of the CubeSat.

A central socket for the Bharat Pi microcontroller, standardized pin headers for sensor connections, and low trace crossovers because to the dual-layer design are all features of the finished PCB.



Key Components of PCB

- TCS34725: RGB and Clear Light Sensor

The TCS34725 is an advanced digital colour sensor capable of detecting red, green, blue, and clear light intensity. It features a built-in IR filter making it easy to integrate with micro-controllers like the Bharat Pi.

In the LUMISAT CubeSat, the TCS34725 plays a critical role in measuring the intensity and colour composition of artificial light in the atmosphere.

This data helps in:

Identifying light pollution patterns

Monitoring spectral composition (e.g., determining if sources are predominantly blue/white LEDs or warmer sodium lamps)

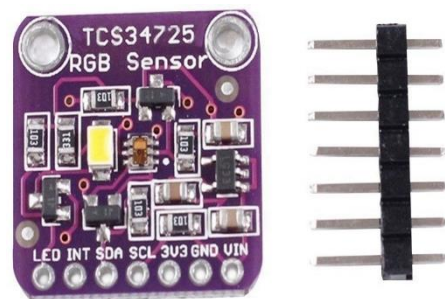
Supporting environmental analysis by correlating colour temperature with ecological impacts.

The sensor has a 4-channel detector array (Red, Green, Blue, and Clear), each capturing light at different wavelengths.

The clear channel provides overall light intensity, useful for measuring lux or ambient brightness.

The built-in IR blocking filter ensures more accurate colour sensing, especially under natural and artificial light sources

It operates on 3.3V or 5V, making it compatible with standard microcontroller logic levels.

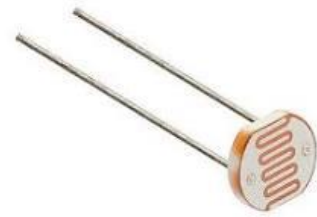


- Photodiode: Light Intensity Detection

A photodiode is a semiconductor-based light sensor that converts incident light into an electrical current. The photodiode is used to:

- Measure the total ambient light intensity (irrespective of colour)
- Support the TCS34725 by providing redundant and confirmatory lux readings
- Detect subtle fluctuations in natural vs. artificial lighting conditions
- Assist in mapping light pollution levels with higher temporal resolution.

The photodiode is connected to one of the analogue input pins of the Bharat Pi.



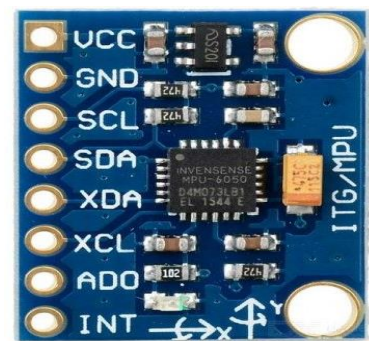
- MPU6050: Accelerometer and Gyroscope Module

The MPU6050 is a 6-axis motion tracking device that combines a 3-axis accelerometer and a 3-axis gyroscope in a single compact module.

In the LUMISAT CubeSat, the MPU6050 is used for:

- Attitude determination: detecting the satellite's orientation in space (pitch, roll, yaw)
- Motion monitoring: tracking any undesired vibrations or rotational motion during flight and descent
- Drop and deployment validation: analysing dynamic forces during launch, free fall, and parachute deployment.

Requires 3.3V to 5V power supply, and it is compatible with the logic levels of Bharat Pi.



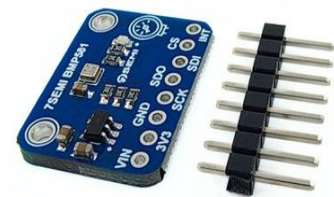
- BMP581: Barometric Pressure and Temperature Sensor

The BMP581 is a high-precision barometric pressure and ambient temperature sensor. It is specifically designed for applications requiring precise altitude measurements, environmental sensing.

In the LUMISAT CubeSat, the BMP581 is used to:

- Measure atmospheric pressure at various altitudes during ascent and descent
- Estimate altitude through the barometric formula with high accuracy

Power requirement is of typically 3.3V to 5V.



- Li-Po Battery: 7.7V, 2000mAh (Lithium-Polymer Rechargeable Battery)

The 7.7V, 2000mAh battery selected for the LUMISAT project provides a balance between compactness, capacity, and voltage stability for onboard electronics.

The Li-Po battery is the **primary power source** for all onboard systems, including:

Bharat Pi microcontroller

All environmental sensors (light, pressure, motion)

GPS and data logging systems

Communication units and memory storage

Connected to a **voltage regulator or Buck converter** to step down voltage to 5V or 3.3V for sensors and controllers.



- Buck Converter: DC-DC Step-Down Voltage Regulator

A buck converter is a high-efficiency DC-DC voltage regulator that steps down a higher input voltage to a lower, stable output voltage. In the LUMISAT CubeSat, the buck converter is a critical power management component, ensuring that all onboard electronics receive safe and regulated voltages derived from the 7.7V Li-Po battery.

Input Voltage: 6V to 12V range (from the Li-Po battery, typically 7.7V)



- SG90 Micro Servo Motor

The SG90 micro servo is a compact, lightweight, and low-power actuator ideal for CubeSats due to its small size, low weight (~9g), and 180° movement range. In LUMISAT, the SG90 servo motor is used for: Deploying the parachute at a predetermined altitude or time.

The servo is powered via the **5V regulated output** from the buck converter, and its **GND is tied** to the common ground with the rest of the system.



- ESP32 S3 DevKitM-1

The ESP32-S3-DevKitM-1 is a tiny, low-power development board based on Espressif's ESP32-S3 SoC, perfect for embedded and IoT applications. It has a dual-core Xtensa LX7 CPU, Bluetooth 5 (LE) and Wi-Fi built in, and sophisticated security features. Without the need for an external programmer, programming and debugging are made simple by its onboard USB interface. The board is appropriate for lightweight data processing on satellites like CubeSats, where effective communication and low power consumption are essential, because it provides AI acceleration for edge computing. Connecting to sensors and modules frequently found in space payloads is made possible by its GPIOs.



TRANSMISSION CODE

```
#include "WiFi.h"
#include "Firebase_ESP_Client.h"
#include "time.h"
#include "Adafruit MPU6050.h"
#include "Adafruit_Sensor.h"
#include "SPI.h"
#include "SparkFun_BMP581_Arduino_Library.h"
#include "ESP32Servo.h"

// Wi-Fi credentials
#define WIFI_SSID "hey"
#define WIFI_PASSWORD "shalu789"

// Firebase credentials
#define API_KEY "AIzaSyB2n9fxEQF4N5wBsSQsdz1VcZPHcZlOpG4"
#define DATABASE_URL "https://lumi-sat-default-rtdb.asia-southeast1.firebaseio.com"
#define USER_EMAIL "orbitalascend@gmail.com"
#define USER_PASSWORD "OrbitalAscend@2025"

// Firebase objects
FirebaseData fbdo;
FirebaseAuth auth;
FirebaseConfig config;
```

```

// MPU6050 object
Adafruit_MPU6050 mpu;

// BMP581 object and SPI pins
BMP581 bmp;
#define SCK_PIN 36
#define MISO_PIN 37
#define MOSI_PIN 35
#define CSN_PIN 34
#define LDR_PIN 4 // Use GPIO1 or another ADC pin

Servo myServo; // Create a Servo object

// Standard sea-level pressure in Pascals
const float SEA_LEVEL_PRESSURE = 101325.0;

void setup() {
  Serial.begin(115200);
  Wire.begin(8, 9); // I2C for MPU6050

  myServo.attach(2); // Attach the servo to GPIO 2

  pinMode(LDR_PIN, INPUT);

  // Initialize MPU6050
  if (!mpu.begin(0x68, &Wire)) {
    Serial.println("MPU6050 init failed ❌");
    while (1) delay(10);
  }
  Serial.println("MPU6050 initialized ✅");
  mpu.setAccelerometerRange(MPU6050_RANGE_8_G);
  mpu.setGyroRange(MPU6050_RANGE_500_DEG);
  mpu.setFilterBandwidth(MPU6050_BAND_21_HZ);

  // Initialize BMP581 over SPI
  SPI.begin(SCK_PIN, MISO_PIN, MOSI_PIN, CSN_PIN);
  while (bmp.beginSPI(CSN_PIN, 100000) != BMP5_OK) {
    Serial.println("BMP581 not connected ❌");
    delay(1000);
  }
  Serial.println("BMP581 connected ✅");

  // Connect to Wi-Fi
  Serial.print("Connecting to Wi-Fi...");
  WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
  while (WiFi.status() != WL_CONNECTED) {
    Serial.print(".");
    delay(300);
  }
  Serial.println(" connected!");

  // Firebase setup
  config.api_key = API_KEY;
  config.database_url = DATABASE_URL;
  auth.user.email = USER_EMAIL;
  auth.user.password = USER_PASSWORD;
  Firebase.begin(&config, &auth);
  Firebase.reconnectWiFi(true);
}

```

```

// Time sync
configTime(0, 0, "pool.ntp.org", "time.nist.gov");

// Clear node on startup
if (Firebase.ready()) {
  if (Firebase.RTDB.deleteNode(&fbdo, "lumisat/data")) {
    Serial.println("Firebase node cleared ✅");
  } else {
    Serial.println("Node clear failed ❌: " + fbdo.errorReason());
  }
}

void loop() {
  struct tm timeinfo;
  String timestamp = "N/A";
  sensors_event_t accel, gyro, temp;
  bmp5_sensor_data bmpData = { 0 };

  mpu.getEvent(&accel, &gyro, &temp);
  bmp.getSensorData(&bmpData);

  if (getLocalTime(&timeinfo)) {
    char buf[20];
    strftime(buf, sizeof(buf), "%H:%M:%S", &timeinfo);
    timestamp = String(buf);
  }

  // Calculate altitude using barometric formula
  float pressure = bmpData.pressure;
  float altitude = 44330.0 * (1.0 - pow(pressure / SEA_LEVEL_PRESSURE, 0.1903));

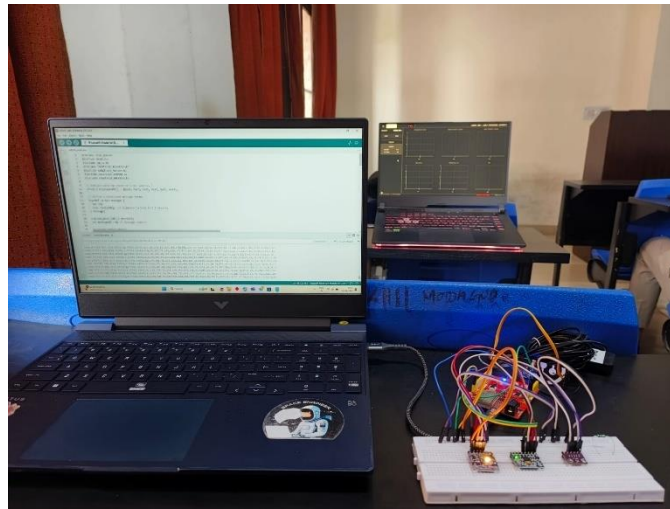
  int lightValue = analogRead(LDR_PIN); // Read LDR value (0-4095 on ESP32)
  Serial.println("LDR: " + String(lightValue));

  if (Firebase.ready()) {
    FirebaseJson json;
    json.set("t", timestamp);
    json.set("ax", accel.acceleration.x);
    json.set("ay", accel.acceleration.y);
    json.set("az", accel.acceleration.z);
    json.set("gx", gyro.gyro.x);
    json.set("gy", gyro.gyro.y);
    json.set("gz", gyro.gyro.z);
    json.set("tmp", bmpData.temperature);
    json.set("p", pressure);
    json.set("alt", altitude); // 📍 Altitude added
    json.set("li", lightValue);

    if (Firebase.RTDB.pushJSON(&fbdo, "lumisat/data", &json)) {
      Serial.println("Uploaded 📁:");
      Serial.println(json.raw());
    } else {
      Serial.println("Upload Failed ❌: " + fbdo.errorReason());
    }
  }

  delay(50);
}

```



Mechanical and Electrical Assembly

The CubeSat structure, 3D printed using PLA with a gyroid infill pattern, was assembled by screwing each face plate onto the vertical support columns. Standard M3 screws were used for precise and rigid fastening. The column design included pre-embedded mounting holes to accommodate each structural panel. The base plate served as the foundation for attaching internal subsystems such as the PCB and power system. The parachute plate was fitted at the top, with precise alignment to allow successful deployment via servo actuation.

The power subsystem, consisting of a 7.7V 2000mAh Li-Po battery and a buck converter, was installed first. The buck converter regulated the voltage to supply the required 5V and 3.3V outputs to different components.

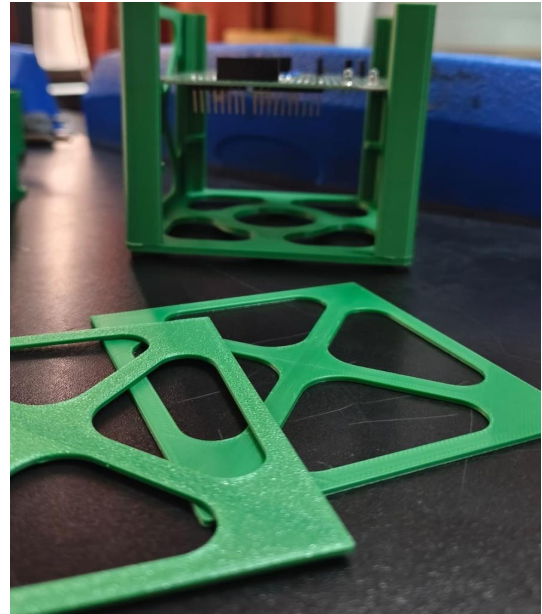
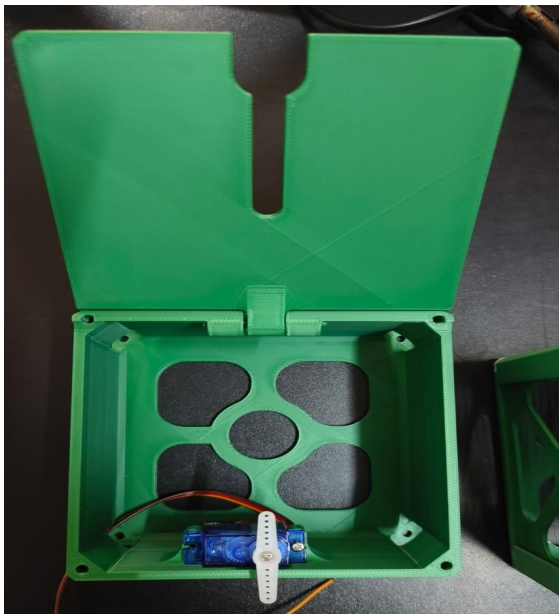
Next, the PCB (8 cm × 8 cm double-layered board) was secured to the base plate using standoffs. It housed the Bharat Pi microcontroller, to which the following components were interfaced:

TCS34725 colour sensor and photodiode for light detection

MPU6050 for inertial data

BMP581 for pressure and temperature sensing

SG90 Servo Motor for actuating the parachute mechanism



Parachute Deployment and Descent Stability Testing

Preparation of Strings for the Parachute: Equal string lengths are manually measured carefully. The strings are securely connected to the specific holes of the CubeSat's columns (presumably on the four edges or corners). This will ensure the parachute is uniformly deployed during the fall.

Parachute Attachment: Once the strings have been measured and attached, the parachute (Ripstop Nylon) is then attached to the CubeSat through these strings. The parachute has to be folded in a special manner to deploy smoothly once the CubeSat has achieved a specific altitude.

Test Deployment: A test drop is performed where the CubeSat (or test model) is dropped from a height that is known. This test is used to confirm the effectiveness of the parachute in deploying correctly and reducing the descent rate. The deployment can be observed to verify that the strings are evenly spaced, and the parachute deploys as expected.

Assembled Model



Ground Station: Real-Time Data Visualization via Web Serial API

To enable real-time reception and visualization of satellite telemetry, a **custom web-based ground station interface** was developed using **JavaScript** and the **Web Serial API**. This interface connects directly to the onboard microcontroller (Bharat Pi) through a **USB-to-serial connection**, allowing ground operators to monitor data without needing third-party software



Link for the Ground station :

<https://lumi-sat.web.app/>

RESULT

The LUMISAT CubeSat successfully demonstrated and verified a multisensory payload for the detection, measurement, and geo-referencing of artificial light in the atmosphere. The outcome includes data acquired from onboard sensors such as the TCS34725 colour sensor, photodiode, BMP581 MPU6050, and ESP32-CAM, all controlled through the Bharat Pi microcontroller and powered by a regulated Li-Po battery system.

One of the key successes was the correct measurement of RGB colour light with the TCS34725, which logged raw red, green, and blue light intensity values along with total luminosity and correlated colour temperature (CCT). This provided greater insight into the quality and intensity of light, which are key parameters for source identification and classification of artificial lighting. These values were logged continuously during the mission, creating a temporal and spatial dataset.

The photodiode was used in parallel to provide binary insights into the presence or absence of light, supplementing the higher resolution data of the TCS34725. By synchronizing the outputs of both light sensors, we created a more reliable and accurate detection system. This combination enhanced the detection of lighting conditions, particularly during transition phases like twilight or where the light intensity fluctuated rapidly.

Positional and environmental information were captured with the BMP581 sensor. These sensors offered real-time altitude, pressure, and geolocation information, enabling us to label the captured light data with accurate coordinates and altitude levels. This geospatial tagging became instrumental during subsequent phases of data comparison and visualization.

The ESP32-CAM provided image frames in low resolution, which were mapped to sensor data for providing visual assurance of lighting conditions. The camera output provided an added layer of interpretability to numerical data so that it could be assured if readings were from artificial sources (streetlights, buildings) or natural sources (sunlight).

All sensor data were transmitted in real time using the Bharat Pi's Wi-Fi module to a JavaScript-based website, where a live dashboard displayed RGB values, altitude, location, and temperature. This streamlined user interface allowed ground station operators to visually assess lighting trends as the CubeSat moved.

In general, the mission confirmed the CubeSat's capability to identify artificial light in diverse environments and altitudes. The quality of the data was sufficient to enable future simulations, classification, and environmental analysis for light pollution.

DISCUSSION

The results of the LUMISAT CubeSat mission provide encouraging implications for environmental monitoring, specifically aimed at artificial light pollution, an under-emphasized but significant source of ecological disturbance and energy loss.

The findings show that the TCS34725 colour sensor, when synchronized with the photodiode, can be employed as an efficient two-layered system for measuring not just the presence of light but also its spectral quality. The dual-sensor system provides greater confidence in the data and allows distinguishing between natural and artificial sources, a feature critical to mapping actual light pollution zones.

With the BMP581 , we were able to attribute each light reading to a certain altitude and geographical coordinate, making a 3D mapping of light intensity possible in the atmosphere. This is a huge improvement from standard 2D light pollution maps. On some trials, it was noted that urban areas had greater RGB values at lower altitudes, while rural areas registered much lower readings. These findings can now be translated into visual graphs and simulation models using software such as MATLAB or AGI's Systems Tool Kit (STK).

By bringing in the sensor data on these platforms, we can visualize comparative light intensity between zones (urban versus rural), monitor the variation with respect to altitude, and even model the likely spread and effects of light pollution based on different atmospheric conditions. This paves the way for more predictive and preventive research on how to control lights in cities in order to reduce ecological footprint and energy loss.

The other important topic of discussion is the ground station integration, where data was shown in real-time through a web-based interface. This interaction demonstrates the feasibility of remote monitoring and even deploying low-Earth monitoring constellations for the continuous tracking of the environment.

Finally, this project also emphasizes the ways in which affordable, modular CubeSats can make significant contributions to environmental science, especially in developing countries where space missions have historically been costly. The LUMISAT design can be replicated for other environmental measurements such as temperature distribution, air quality, or even atmospheric particle monitoring.

CONCLUSION

The LUMISAT CubeSat mission is an important milestone in proving the efficacy of low-cost, student-designed satellite systems to be applied to environmental monitoring in the form of light pollution detection and its overall impact on sustainability and urban design. By demonstrating the successful fusion of hardware, software, and simulation techniques, the mission has given insight into the engineering as well as the scientific aspects of remote sensing application.

The CubeSat structural design, created with Fusion 360 and SolidWorks, provided the best possible balance between weight and rigidity, and the application of PLA material was proven through ANSYS-based structural and drop tests. The ultimate payload structure was found to withstand simulated stress conditions, proving it ready for deployment at high altitudes. The deployment system utilizing a 1.2-meter ripstop nylon parachute, complemented by calculations of descent rate from drag coefficient theory, further improved the feasibility of controlled descent, with recovery of onboard electronics ensured.

The onboard sensor cluster, the TCS34725 colour sensor, photodiode, BMP280 barometric sensor, MPU6050 IMU, and NAVIC GPS module was also interfaced successfully with the Bharat Pi microcontroller. These sensors collectively offered rich, real-time environmental information such as RGB colour intensity, temperature, pressure, orientation, and accurate geolocation. This allowed for extensive geotagged mapping of artificial light sources. The inclusion of a camera module (ESP32-CAM) provided visual context to raw sensor data, improving interpretability and reliability of results.

One of the fundamental contributions of the project is the efficient transfer of sensor information to a JavaScript-built web-based ground station and Web Serial API. This enabled real-time visual data of satellites on a computer monitor to monitor flight behaviour and light pollution levels in real-time. The employment of NAVIC GPS, India's domestic satellite navigation system not only enhanced positional accuracy within the country but also proved self-reliant space technology integration's potential in academic research.

Additionally, the project successfully implemented a robust power management system using a 7.7V 2000 Mah Li-Po battery regulated by a buck converter, ensuring adequate power supply throughout the mission duration. This power subsystem demonstrated efficiency and reliability, vital for long-duration environmental sensing.

In summary, LUMISAT illustrates the capabilities of a relatively small-sized and affordable CubeSat as a highly useful tool for solving practical problems like artificial light pollution and environmental consequences. Success in the project provides avenues for additional research into urban sustainability, space-based environmental science, and technology democratization. It also showcases the promise for student-led innovation to make meaningful contributions to national and international issues, opening doors for future missions of higher complexity and larger scientific ambitions.

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